DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2025-26
PRE BOARD EXAMINATION

CLASS: X **FULL MARKS: 80**
SUBJECT: CHEMISTRY [SET A] **TIME: 2 HOURS**

Maximum Marks: 80 Time allowed: Two hours
Answers to this Paper must be written on the paper provided separately. You will not be allowed to write during first 15 minutes.
This time is to be spent in reading the question paper.
The time given at the head of this Paper is the time allowed for writing the answers.

Section A is compulsory. Attempt any four questions from Section B. The intended marks for questions or parts of questions are given in brackets [ ].
(This paper consists of seven printed pages)

SECTION A
(Attempt all questions from this Section)

Question 1
Choose the correct answers to the questions from the given options. **[15]**
(Do not copy the questions, write the correct answers only.)

(i) A compound 'X' on electrolysis in its aqueous solution produces a metal at the cathode which on reaction with water produces a strong alkali and a greenish yellow gas at the anode. 'X' is:
(a) Magnesium chloride **(c) Cupric chloride**
(b) Sodium chloride **(d) Zinc sulphate**

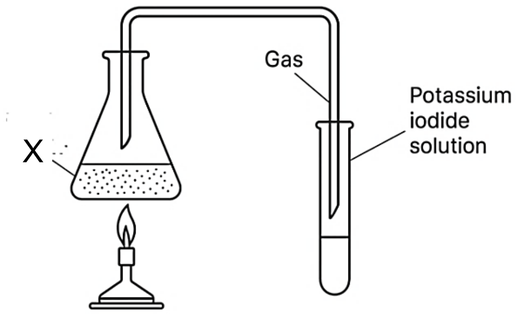
(ii) An acid 'A' is treated with silver nitrate solution to obtain a curdy white ppt. Then it is treated with ammonium hydroxide solution when the ppt. dissolves. Identify 'A'
(a) Sulphuric acid **(c) Hydrochloric acid**
(b) Acetic acid **(d) Nitric acid**

(iii) Assertion (A): Potassium chloride conducts electricity in aqueous solution
Reason (R): Ions are free to move in molten state or in dissolved state
(a) (A) is true and (R) is false
(b) (A) is false and (R) is true
(c) Both (A) and (R) are true and R is the correct explanation of (A)
(d) Both (A) and (R) are true, but R is not the correct explanation of (A)

(iv) Sodium atom and sodium ions:
(a) are chemically same
(b) have same number of electrons
(c) have same number of protons
(d) none of the above

(v) The pH values of four solutions P, Q, R and S are 1, 7, 8 and 13 respectively. Which solution will require the maximum volume of sodium hydroxide solution for complete neutralization, assuming all have same molar concentration and volume?
(a) P **(b) Q** **(c) R** **(d) S**

(vi) **A compound 'X' is heated in a conical flask. The gas liberated turns potassium iodide solution brown. The liberated gas could possibly be?**

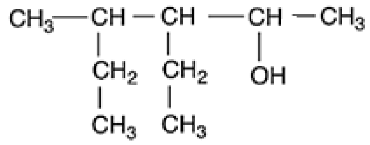


(a) Oxygen **(b) Chlorine** **(c) Sulphur dioxide** **(d) Nitrogen dioxide**

(vii) **The atomic masses of oxygen, fluorine and neon are 16, 19 and 20 respectively. Which of the following statements regarding the volume of gases at STP in 16g of oxygen, 19 g of fluorine and 20 g of neon is correct?**
M: 16 g of oxygen will occupy half the volume occupied by 20 g of neon
N: 20g of neon will occupy twice the volume occupied by 19 g of fluorine
(a) Both M and N **(c) Only N**
(b) Neither M nor N **(d) Only M**

(viii) **An element 'X' has electronic configuration 2, 8, 6. It's oxide when dissolved in water will:**
(a) turn red litmus blue **(c) turn blue litmus red**
(b) have no effect on litmus **(d) liberate hydrogen gas**

(ix) **The structure of an organic compound is given below. What is its correct IUPAC name?**



(a) 3, 4-diethyl-4-methylbutan-2-ol **(c) 3, 4-diethylpentan-2-ol**
(b) 2, 3-diethylpentan- 2-ol **(d) 3-ethyl-4-methylhexan-2-ol**

(x) **A negative divalent element will belong to which group?**
(a) Group 2 **(b) Group 13** **(c) Group 14** **(d) Group 16**

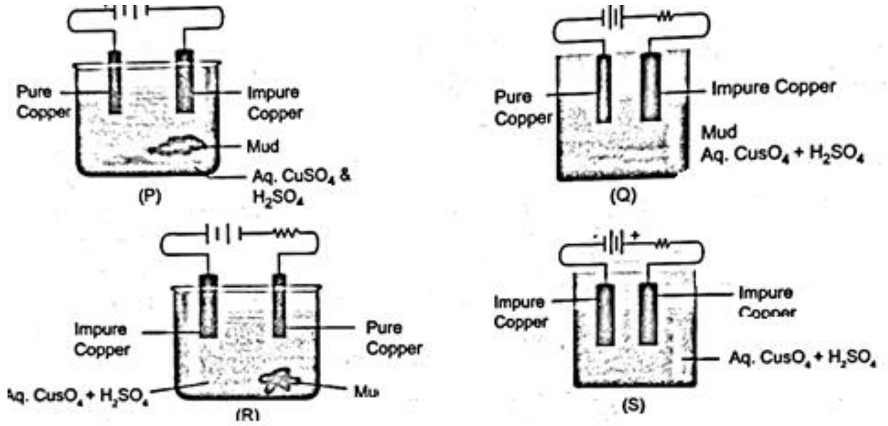
(xi) **The ratio of the number of oxygen atoms in 9.8g of H_2SO_4 to the number of hydrogen atoms in 0.6g of CH_4 is:** **[S = 32, C = 12, O = 16]**
(a) 1:1 **(b) 2:1** **(c) 1:2** **(d) 8:3**

(xii) **Electrolysis of molten lead bromide liberates reddish brown fumes at the anode. Which of the following is the most appropriate option for this electrolysis?**
(a) Graphite electrodes and silica crucible **(b) Platinum electrode and glass crucible**
(c) Graphite electrodes and glass crucible **(d) Platinum electrode and silica crucible**

(xiii) **– CHO is the functional group present in:**
(a) Butanal **(b) Ethanol** **(c) Propanone** **(d) Methanoic acid**

(xiv) **The empirical formula of an organic compound is $C_3H_6O_2$ and the vapour density of the compound is 74. Find its molecular formula [C=12, O=16]**
(a) $C_3H_6O_2$ **(b) $C_6H_8O_2$** **(c) $C_6H_4O_2$** **(d) $C_6H_{12}O_4$**

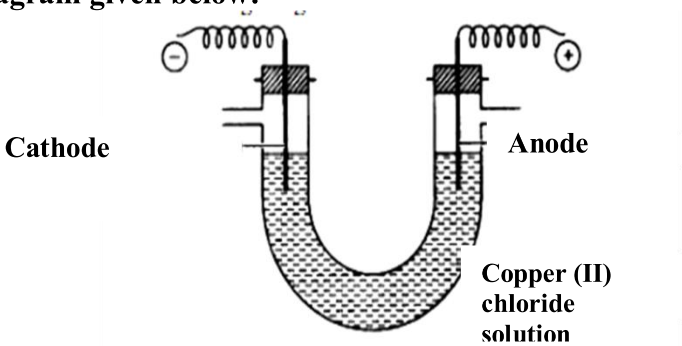
(xv) **Which of the following electrolytic cell [P, Q, R or S] represents the electrolysis of copper sulphate correctly?**



(a) P (b) Q (c) R (d) S

Question 2

(i) A solution of aqueous copper (II) chloride is electrolysed using graphite electrodes as illustrated in the diagram given below: **[5]**



(a) Name the ions which will migrate to the anode.

(b) Which ion is likely to discharge at the cathode and why?
(c) Write the ionic equation taking place at the anode.
(d) Name the product formed at the cathode.

(ii) Identify the following: **[5]**
(a) A salt formed by the partial replacement of hydroxyl ions of a base by an acid.
(b) Aliphatic, open chain organic compounds containing carbon and hydrogen only.
(c) The naturally occurring minerals from which metals can be extracted profitably and conveniently.
(d) A homogenous mixture of two or more metals or a metal and a nonmetal in a definite proportion in their molten state.
(e) The electrolyte used in electroplating a key with silver.

(iii) Complete the following by choosing the correct answers from the bracket: **[5]**
(a) If an element has seven valence electrons, then it is likely to have the ____ (smallest/ highest) electron affinity among all the elements in the same period.
(b) The gas produced when ethanol is heated with conc. sulphuric acid at 170^{O}C is ____ (ethene/ ethane).
(c) A bond formed between two atoms where the shared pair of electrons is contributed by only one of the atoms is ___ (coordinate/ ionic).

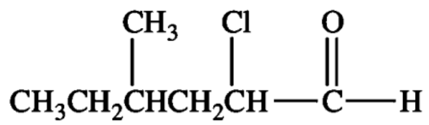
(d) An _____ (acidic/ alkaline) solution will turn alkaline phenolphthalein colourless.

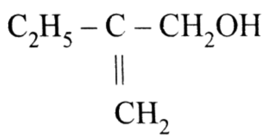
(e) Hydrochloric acid is reacted with sodium sulphide to liberate a gas which turns moist lead acetate paper _______ (black/yellow).

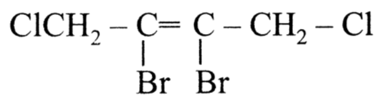
(iv) Match the items in Column A with the most appropriate ones in column B **[5]**

Column A	Column B
(a) Froth flotation	1. Ionic
(b) Calcination	2. Dehydrating agent
(c) Calcium oxide	3. Sulphide ore
(d) Nitric Acid	4. Carbonate ore
(e) Conc. Sulphuric acid	5. Strong oxidizing agent

(v) (a) Give the IUPAC names of the following compounds: **[3]**

1. 

2. 

3. 

(b) Draw the structural formula of the following organic compounds: **[2]**

1. 2-methylpropanoic acid
2. Acetaldehyde

SECTION B (40 Marks)

(Attempt any four questions)

Question 3

(i) Give one significant observation for each of the following: **[2]**

(a) Sodium is added to ethanol.

(b) Dry ammonia gas is passed over heated lead oxide placed in a combustion tube.

(ii) Give reasons for the following statements: **[2]**

(a) Pure acetic acid is called glacial acetic acid.

(b) Ethene undergoes addition reaction.

(iii) An element 'X' (2, 8, 7) belongs to 17^{th} group of the periodic table. Based on this information, complete the following: **[3]**

(a) The electronic configuration of its ion is ____.

(b) It combines with group 1 and group 2 elements by ____ bonding.

(c) The electron affinity of fluorine is ____ than that of 'X'.

(iv) Write balanced chemical equations for the following: **[3]**

(a) Catalytic oxidation of sulphur dioxide.

(b) Addition of cold water to calcium carbide.

(c) Excess chlorine is passed through ammonia.

Question 4

(i) Name the main metal present in the following alloys: **[2]**

(a) Duralumin **(b) Stainless steel**

(ii) Write balanced chemical equations for the following: **[2]**

(a) Preparation of ethene from ethanol using a concentrated acid.

(b) Laboratory preparation of nitric acid from a non-volatile acid.

(iii) Seema takes a blue crystalline salt P in a test tube. On heating it produces a white anhydrous powder. P is dissolved in water. Zinc is added to one part of the solution and to another part of the solution barium chloride solution is added. **[3]**

(a) Name the compound P.

(b) Mention one observation when zinc is added to the solution of P.

(c) State the colour of the precipitate formed when barium chloride is added to the solution of P.

(iv) A compound R, which has a sharp vinegar like odour reacts with ethanol in the presence of a mineral acid to form a sweet-smelling compound Q and water. **[3]**

(a) Name the type of reaction taking place.

(b) Write a balanced chemical equation for the reaction.

(c) State the function of mineral acid in the above reaction.

Question 5

(i) Identify the reactant and write the balanced equation for the following: **[2]**

Nitric acid reacts with compound Q to give a salt $Ca(NO_3)_2$, water and carbon dioxide.

(ii) Determine the empirical formula of a compound containing 47.9% K, 5.5% Be and 46.6% fluorine by mass. [Be=9, F=19, K=39] **[2]**

(iii) State the property exhibited by sulphuric acid in each of the following reactions: **[3]**

(a) As a source of hydrogen when treated in dilute form with metals like Zn, Mg, Fe etc.

(b) Charring of sugar with hot concentrated acid.

(c) Production of hydrogen chloride on treating concentrated acid with sodium chloride.

(iv) Give balanced equations for the following: **[3]**

(a) Preparation of ethyne from ethylene dibromide.

(b) Incomplete combustion of ethane.

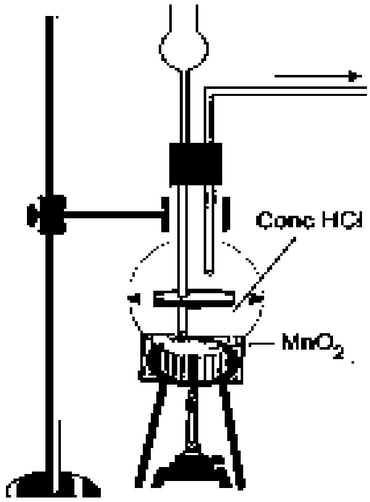
(c) Bromine gas is passed over ethene in presence of carbon tetrachloride.

Question 6

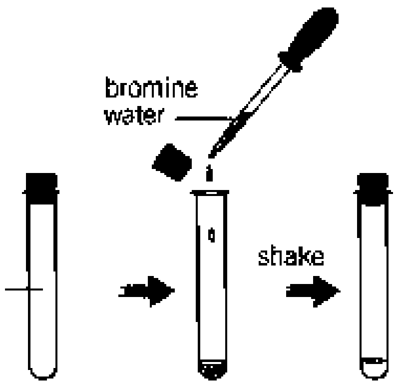
(i) A student was asked to perform two experiments in the laboratory based on the instructions given: **[2]**

Observe the picture given below and state one observation for each of the Experiments 1 and 2 that you would notice.

(a) **Experiment 1**



(b) **Experiment 2**



(ii) You are provided with the list of chemicals mentioned below in the box: **[2]**

copper, iron, nitre, conc. sulphuric acid, dilute nitric acid, dil. hydrochloric acid, dil. sulphuric acid, nitrogen, water

Using suitable chemicals from the list given, write balanced chemical equation for the preparation of the compounds mentioned below:
(a) Copper nitrate
(b) Nitric acid vapours

(iii) Using the equation given below, calculate the following: **[3]**
$2NH_4Cl + Ca(OH)_2 \rightarrow CaCl_2 + 2H_2O + 2NH_3$
(a) Volume of ammonia at STP obtained from 4.28g of ammonium chloride.
(b) Weight of calcium hydroxide needed to decompose 2.14g of ammonium chloride.
(c) Moles of water evolved from 2.14g of ammonium chloride.
[N=14, Ca=40, Cl=35.5]

(iv) Identify P, Q and R in the following reaction. **[3]**
(a) $P + 2HCl \rightarrow MgCl_2 + H_2O + CO_2$
(b) $4NH_3 + 3O_2 \rightarrow Q + 6H_2O$
(c) $R + \text{sodalime} \rightarrow C_2H_6 + Na_2CO_3$

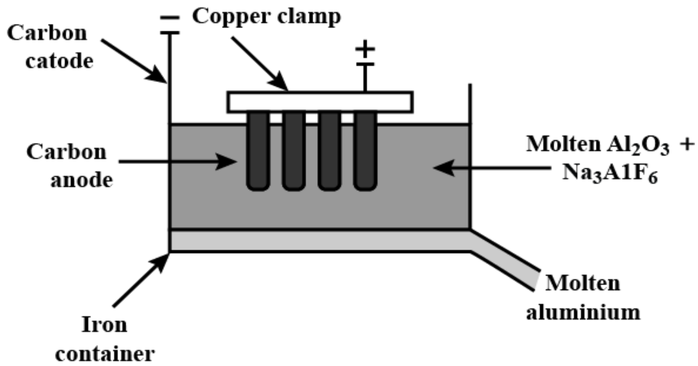
Question 7

(i) Give reasons for the following: [2]

(a) Silver-plated cutlery is not considered as pure silver.

(b) Concentrated nitric acid appears yellow when it is left standing in a glass bottle.

(ii) Given below is a schematic diagram of the electrolytic reduction of alumina. [2]



(a) Why cryolite is added to molten alumina in the electrolytic tank?

(b) Write the reaction taking place at the anode.

(iii) Give balanced equations for each of the following: [3]

(a) Action of warm water on magnesium nitride.

(b) Oxidation of sulphur with conc. nitric acid.

(c) Laboratory preparation of ethanol by using chloroethane and aqueous sodium hydroxide.

(iv) Amit has solutions X, Y and Z having pH 4, 7 and 10 respectively. Which solution will: [3]

(a) liberate carbon dioxide when reacted with sodium bicarbonate?

(b) turn methyl orange yellow?

(c) dissolve zinc metal to liberate hydrogen gas?

Question 8

(i) State giving reasons if: [2]

(a) Zinc and aluminium can be distinguished by heating the metal powder with concentrated sodium hydroxide solution.

(b) Calcium nitrate and lead nitrate can be distinguished by adding ammonium hydroxide solution to the salt solution.

(ii) Draw the electron dot diagram of ammonium ion. [2]

(iii) Write balanced chemical equation for the following conversions (A to C): [3]

$$\text{Lead} \xleftarrow{A} \text{Lead(II) oxide} \xrightarrow{B} \text{Lead(II) nitrate} \xrightarrow{C} \text{Lead(II) hydroxide}$$

(iv) L, M and N are three elements with atomic numbers 12, 15 and 17 respectively. [3]
Answer the following questions using only the alphabets given.
Do not identify the elements.

(a) Which element forms a divalent cation?

(b) Which element has highest electronegativity?

(c) Which two elements will form a covalent compound?